SUBJECT: Pre-Shipment Returns Risk Model & Intervention Strategy

1. The Decision

Call flagged orders immediately—do not hold dispatch.

Our sensitivity and margin analysis reveals that holding orders for more than 24 hours is **economically negative** across all product categories. Holding creates friction, leading roughly **12% of legitimate buyers to cancel their orders entirely**, which wipes out any potential logistics savings.

Instead of holding shipments, Kestrel should use the model's risk score to trigger an immediate, proactive phone call / verification workflow while the order remains in the standard dispatch queue.

2. The Number

While a target of "95% accuracy" sounds compelling, raw accuracy on imbalanced return data is misleading (a model predicting "No Return" on every order achieves ~90% accuracy while saving ₹0).

Our evaluation focuses on **Precision @ Top Decile** and **Risk Score Calibration**:

- **Target Capture:** The model flags the top high-risk orders where intervention yields the highest ROI.
- **Operational Threshold:** By setting a high risk-score threshold, we only route orders to the outbound call team when the expected return cost exceeds the cost of making the call.

3. The Rupees (Financial Trade-off)

- **The Cost of Holding:** For every 100 orders held past 24 hours, ~12 orders are canceled. On high-margin appliances (mixers, air fryers, robot vacuums), losing those 12 sales hurts gross margin far more than absorbing a few return shipping fees.
- **The Math of Calling:** An outbound confirmation call costs a few rupees. Preventing a single return saves **₹180–₹350** in forward/reverse freight and repackaging costs.
- **Breakeven Target:** The call intervention pays for itself if it prevents at least **~15% of returns** among flagged high-risk orders.

4. What You Should Do Next Week

Launch a **2-week controlled pilot** across approximately **2,000 flagged high-risk orders**:

1. **50-50 Split Trial:**
    - **Group A (1,000 orders):** Outbound confirmation call placed prior to dispatch.
    - **Group B (1,000 orders):** Control group (standard dispatch, no call).

2. **Key Success Metric:** Measure whether Group A shows a **$\ge 15\%$ reduction in actual returns** compared to Group B without driving up customer cancellation rates.
3. **Rollout Trigger:** If the 15% threshold is met, scale the intervention permanently across all D2C lines.